

PostgreSQL Backup & Restore to New Host with WAL Archive files

How to Backup & Restore + Recover a PostgreSQL to New Host using pg_basebackup+WAL Archives

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0. Goal

Perform a full recovery to the latest consistent state by restoring a complete backup taken with **pg_basebackup** and applying the **WAL archive files**.

Note: pg_basebackup does **not** include WAL archive files (archive log files) in its backup; these must be backed up separately.

We have a base backup taken **without using –tablespace-mapping**, and the restore needs to be performed on a different host with the following differences:

- Tablespace mount point paths are different from the source server.
- Archive WAL directory is located in a different path.
- WAL file (pg_wal) location differs.
- PGDATA directory path is also different.

1. Environment (Source / Target)

ASPECT	SOURCE	TARGET	DI
Hostname	lxicbpgdsgv01	lxicbpgdsgv02	Dif hos
IP Address	192.168.2.51	192.168.2.52	Dif
OS	RHEL 9	RHEL 9	Sar
DB Version	PostgreSQL v17.6	PostgreSQL v17.6	Sar
Archive mode	archive_mode=on	archive_mode=on	Sar

ASPECT	SOURCE	TARGET	DI
PGDATA	/pgData/pgsql17/data	/pgdata/pgsql17/data	Difi cas
WAL Directory	/pgWal/pgsql17/wal	/pgwal/pgsql17/wal	Difi cas
WAL Archive Directory	/pgArch/pgsql17/arch	/pgarchive/pgsql17/archive	Difi cas
Backup Directory	/pgBackup/pgsql17/backup	/pgbackup/pgsql17/backup	Difi cas
restore_command	Not Applicable	/pgbackup/pgsql17/backup/archive_bkp11OCT2025	res on for usi file
Tablespace & Path	DELL_TB_DATA:/pgTb/pgsql17/tbs/delltbs01 ORCL_TB_DATA:/pgTb/pgsql17/tbs/orcltbs01	DELL_TB_DATA:/pgtsdata01/pgsql17/tbs/delltbs01 ORCL_TB_DATA:/pgtsdata01/pgsql17/tbs/orcltbs01	It v ing ble but loc
Databases	DELL, ORCL	We will restore both from backup	We bot

2. Verify Existing DB Setup

```
[postgres@lxicbpgdsgv01 ~]$ psql
psql (17.6)
Type "help" for help.

postgres=# \l+

List of databases
  Name      | Owner      | Encoding | Locale Provider | Collate   | Ctype     | Locale | ICU
Rules | Access privileges | Size  | Tablespace | Description
```

```

-----+-----+-----+-----+-----+-----+-----+
--+-----+-----+-----+-----+-----+-----+
dell      | postgres | UTF8    | libc      | en_SG.UTF-8 | en_SG.UTF-8 |      |
|          |          |          |           |              |              |      |
|          | 7755 kB | dell_tb_data |          |              |              |      |
orcl      | postgres | UTF8    | libc      | en_SG.UTF-8 | en_SG.UTF-8 |      |
|          | 7569 kB | orcl_tb_data |          |              |              |      |
postgres  | postgres | UTF8    | libc      | en_SG.UTF-8 | en_SG.UTF-8 |      |
|          | 492 MB | pg_default | default administrative connection database
template0 | postgres | UTF8    | libc      | en_SG.UTF-8 | en_SG.UTF-8 |      |
| =c/postgres +| 7545 kB | pg_default | unmodifiable empty database
|          |        |          |          |              |              |      |
| postgres=Ctc/postgres |          |          |
template1 | postgres | UTF8    | libc      | en_SG.UTF-8 | en_SG.UTF-8 |      |
| =c/postgres +| 7569 kB | pg_default | default template for new databases
|          |        |          |          |              |              |      |
| postgres=Ctc/postgres |          |          |
(5 rows)

```

postgres=# \db

```

              List of tablespaces
   Name      | Owner   | Location
-----+-----+-----+
dell_tb_data | postgres | /pgTb/pgsql17/tbs/delltbs01
orcl_tb_data | postgres | /pgTb/pgsql17/tbs/orcltbs01
pg_default   | postgres |
pg_global    | postgres |
(4 rows)

```

postgres=#

postgres=# \c dell

You are now connected to database "dell" as user "postgres".

dell=# CREATE TABLE test.emp (name TEXT, designation TEXT, project TEXT, company TEXT) TABLESPACE DELL_TB_DATA;

CREATE TABLE

dell=# INSERT INTO test.emp VALUES ('Sugi', 'DBA', 'Jetstar', 'iGATE');
INSERT 0 1

dell=# INSERT INTO test.emp VALUES ('Teja', 'DBA', 'RCM', 'iGATE');
INSERT 0 1

dell=# INSERT INTO test.emp VALUES ('RAJ', 'DBA', 'RCM', 'iGATE');
INSERT 0 1

dell=# CHECKPOINT;

CHECKPOINT

dell=# SELECT * FROM TEST.EMP;

```

 name | designation | project | company

```

```

-----+-----+-----+-----
Sugi | DBA          | Jetstar | iGATE
Teja | DBA          | RCM     | iGATE
RAJ  | DBA          | RCM     | iGATE
(3 rows)

```

```
dell=#
```

3. Pre-requisites

3.1. Verify postgresql.conf

```

postgres=# SHOW wal_level;
wal_level
-----
replica <----
(1 row)

postgres=# show archive_mode;
archive_mode
-----
on <----
(1 row)

postgres=# show archive_command;
archive_command
-----
cp %p /pgArch/pgsql17/arch/%f
(1 row)

postgres=#

```

3.2. Verify pg_hba.conf

```

# TYPE      DATABASE          USER            ADDRESS          METHOD
# Local connections for replication (for pg_basebackup run locally)
local      replication    all                                trust

```

```
# Remote connections for replication (for pg_basebackup run remotely)
#host      replication      repl_user      192.168.2.52/32      scram-sha-256
```

3.3. Verify User Permissions (REPLICATION or SUPERUSER)

```
postgres=# \du
                                List of roles
Role name |                               Attributes
-----+-----
postgres | Superuser, Create role, Create DB, Replication, Bypass RLS

postgres=#
```

3.4. Verify Free Space for Backup

```
[postgres@lxicbpgdsgv01 ~]$ df -h /pgBackup
Filesystem      Size  Used Avail Use% Mounted on
/dev/sdd1       100G  746M   100G   1% /pgBackup
[postgres@lxicbpgdsgv01 ~]$
# Create directories for base backup and WAL archive log files
[postgres@lxicbpgdsgv01 ~]$ mkdir -p
/pgBackup/pgsql17/backup/base_bkp11OCT2025
[postgres@lxicbpgdsgv01 ~]$ mkdir -p
/pgBackup/pgsql17/backup/archive_bkp11OCT2025
[postgres@lxicbpgdsgv01 ~]$
```

4. Take Base Backup and WAL Archive

4.1. Take Base Backup

```
# Take base backup

[postgres@lxicbpgdsgv01 ~]$ nohup pg_basebackup -U postgres -D
/pgBackup/pgsql17/backup/base_bkp11OCT2025 -Ft -Xs -P -v >
/pgBackup/pgsql17/backup/base_bkp11OCT2025.log 2>&1 &
[1] 3413
[postgres@lxicbpgdsgv01 ~]$
[postgres@lxicbpgdsgv01 ~]$ cat
/pgBackup/pgsql17/backup/base_bkp11OCT2025.log
nohup: ignoring input
pg_basebackup: initiating base backup, waiting for checkpoint to complete
```

```
pg_basebackup: checkpoint completed
pg_basebackup: write-ahead log start point: 0/8C000028 on timeline 1
pg_basebackup: starting background WAL receiver
pg_basebackup: created temporary replication slot "pg_basebackup_3416"
  7754/535855 kB (1%), 0/3 tablespaces
(...ckup/base_bkp11OCT2025/16542.tar)
  7754/535855 kB (1%), 1/3 tablespaces
(...ckup/base_bkp11OCT2025/16542.tar)
 15474/535855 kB (2%), 1/3 tablespaces
(...ckup/base_bkp11OCT2025/16545.tar)
 15474/535855 kB (2%), 2/3 tablespaces
(...ckup/base_bkp11OCT2025/16545.tar)
157800/535855 kB (29%), 2/3 tablespaces
(...ackup/base_bkp11OCT2025/base.tar)
342056/535855 kB (63%), 2/3 tablespaces
(...ackup/base_bkp11OCT2025/base.tar)
496488/535855 kB (92%), 2/3 tablespaces
(...ackup/base_bkp11OCT2025/base.tar)
535868/535868 kB (100%), 2/3 tablespaces
(...ackup/base_bkp11OCT2025/base.tar)
535868/535868 kB (100%), 3/3 tablespaces
pg_basebackup: write-ahead log end point: 0/8C000120
pg_basebackup: waiting for background process to finish streaming ...
pg_basebackup: syncing data to disk ...
pg_basebackup: renaming backup_manifest.tmp to backup_manifest
pg_basebackup: base backup completed <-----
```

```
[postgres@lxicbpgdsgv01 ~]$
```

```
[postgres@lxicbpgdsgv01 ~]$ du -sh
/pgBackup/pgsql17/backup/base_bkp11OCT2025
540M    /pgBackup/pgsql17/backup/base_bkp11OCT2025
```

```
[postgres@lxicbpgdsgv01 ~]$
```

```
[postgres@lxicbpgdsgv01 ~]$ ls -ltr
/pgBackup/pgsql17/backup/base_bkp11OCT2025
```

```
total 552504
-rw-----. 1 postgres postgres  7940096 Oct 12 00:42 16542.tar
-rw-----. 1 postgres postgres  7905792 Oct 12 00:42 16545.tar
-rw-----. 1 postgres postgres 532883456 Oct 12 00:42 base.tar
-rw-----. 1 postgres postgres   243611 Oct 12 00:42 backup_manifest
-rw-----. 1 postgres postgres 16778752 Oct 12 00:42 pg_wal.tar
[postgres@lxicbpgdsgv01 ~]$
```

backup_manifest is a JSON file included inside the tar backup.

It contains metadata about the backup, like:

--> List of all files included in the backup

--> Their sizes and checksums

```
--> Backup timestamps
--> Backup type (full, incremental, differential)
--> Other metadata needed to verify and restore the backup properly
```

4.2. Insert new row on table test.emp after backup

```
postgres=# \c dell
You are now connected to database "dell" as user "postgres".
dell=# INSERT INTO test.emp VALUES ('KK', 'DBA LEAD', 'RCM', 'iGATE');
INSERT 0 1
dell=# CHECKPOINT;
CHECKPOINT
dell=# select * from test.emp;
   name | designation | project | company
-----+-----+-----+-----
  Sugi  | DBA         | Jetstar | iGATE
  Teja  | DBA         | RCM     | iGATE
  RAJ   | DBA         | RCM     | iGATE
  KK    | DBA LEAD   | RCM     | iGATE <---- New row
(4 rows)

# Switch Archivelog (WAL Archive logs)
dell=# SELECT pg_switch_wal();
 pg_switch_wal
-----
 0/8D000328
(1 row)

dell=# SELECT pg_switch_wal();
 pg_switch_wal
-----
 0/8E000000
(1 row)

dell=# SELECT pg_switch_wal();
 pg_switch_wal
-----
 0/8E000000
(1 row)

dell=#
```


4.3. Copy WAL Archive Files Separately

```
[postgres@lxicbpgdsgv01 ~]$ cp /pgArch/pgsql17/arch/*  
/pgBackup/pgsql17/backup/archive_bkp11OCT2025  
[postgres@lxicbpgdsgv01 ~]$  
[postgres@lxicbpgdsgv01 ~]$ ls -ltr  
/pgBackup/pgsql17/backup/archive_bkp11OCT2025  
total 65540  
-rw-----. 1 postgres postgres 16777216 Oct 12 00:53  
0000000100000000000000008A  
-rw-----. 1 postgres postgres 16777216 Oct 12 00:53  
0000000100000000000000008B  
-rw-----. 1 postgres postgres      341 Oct 12 00:53  
0000000100000000000000008C.00000028.backup  
-rw-----. 1 postgres postgres 16777216 Oct 12 00:53  
0000000100000000000000008C  
-rw-----. 1 postgres postgres 16777216 Oct 12 00:53  
0000000100000000000000008D  
[postgres@lxicbpgdsgv01 ~]$
```

5. Transfer Base Backup and WAL Archives to Target

```
[postgres@lxicbpgdsgv01 ~]$ cd /pgBackup/pgsql17/backup/  
[postgres@lxicbpgdsgv01 backup]$ ls -ltr  
total 4  
drwxr-xr-x. 2 postgres postgres  97 Oct 12 00:42 base_bkp11OCT2025  
-rw-r--r--. 1 postgres postgres 1286 Oct 12 00:42 base_bkp11OCT2025.log  
drwxr-xr-x. 2 postgres postgres 182 Oct 12 00:56 archive_bkp11OCT2025  
[postgres@lxicbpgdsgv01 backup]$  
[postgres@lxicbpgdsgv01 backup]$ tar -cvf base_bkp11OCT2025.tar  
base_bkp11OCT2025  
base_bkp11OCT2025/  
base_bkp11OCT2025/16542.tar  
base_bkp11OCT2025/pg_wal.tar  
base_bkp11OCT2025/16545.tar  
base_bkp11OCT2025/base.tar  
base_bkp11OCT2025/backup_manifest  
[postgres@lxicbpgdsgv01 backup]$  
[postgres@lxicbpgdsgv01 backup]$ tar -cvf archive_bkp11OCT2025.tar  
archive_bkp11OCT2025  
archive_bkp11OCT2025/  
archive_bkp11OCT2025/0000000100000000000000008A  
archive_bkp11OCT2025/0000000100000000000000008B  
archive_bkp11OCT2025/0000000100000000000000008C
```

```

archive_bkp11OCT2025/000000010000000000000008C.00000028.backup
archive_bkp11OCT2025/000000010000000000000008D
[postgres@lxicbpgdsgv01 backup]$
[postgres@lxicbpgdsgv01 backup]$ ls -ltr
total 618056
drwxr-xr-x. 2 postgres postgres          97 Oct 12 00:42 base_bkp11OCT2025
-rw-r--r--. 1 postgres postgres       1286 Oct 12 00:42
base_bkp11OCT2025.log
drwxr-xr-x. 2 postgres postgres        182 Oct 12 00:56 archive_bkp11OCT2025
-rw-r--r--. 1 postgres postgres 565760000 Oct 12 01:00
base_bkp11OCT2025.tar
-rw-r--r--. 1 postgres postgres 67123200 Oct 12 01:01
archive_bkp11OCT2025.tar
[postgres@lxicbpgdsgv01 backup]$

[postgres@lxicbpgdsgv01 backup]$ scp *.tar
192.168.2.52:/pgbackup/pgsql17/backup
postgres@192.168.2.52's password:
archive_bkp11OCT2025.tar          100%   64MB   54.6MB/s   00:01
base_bkp11OCT2025.tar            100%  540MB   58.3MB/s   00:09
[postgres@lxicbpgdsgv01 backup]$

```

6. Check and Prepare Target Environment

6.1. Check Existing symbolic link and Mounts

```

[postgres@lxicbpgdsgv02 ~]$ find /pgdata/pgsql17/data -type l -ls
67117239      0 lrwxrwxrwx   1 postgres postgres      18 Oct 10 19:43
/pgdata/pgsql17/data/pg_wal -> /pgwal/pgsql17/wal
[postgres@lxicbpgdsgv02 ~]$

```

6.2. Create Tablespace Directories

```

[postgres@lxicbpgdsgv02 ~]$ mkdir -p /pgtsdata01/pgsql17/tbs/delltbs01
[postgres@lxicbpgdsgv02 ~]$ mkdir -p /pgtsdata01/pgsql17/tbs/orcltbs01
[postgres@lxicbpgdsgv02 ~]$ chown -R postgres:postgres
/pgtsdata01/pgsql17/tbs/delltbs01
[postgres@lxicbpgdsgv02 ~]$ chown -R postgres:postgres
/pgtsdata01/pgsql17/tbs/orcltbs01
[postgres@lxicbpgdsgv02 ~]$ chmod 700 /pgtsdata01/pgsql17/tbs/delltbs01

```

```
[postgres@lxicbpgdsgv02 ~]$ chmod 700 /pgtsdata01/pgsql17/tbs/orcltbs01
[postgres@lxicbpgdsgv02 ~]$
```

6.3. Stop PostgreSQL

```
[root@lxicbpgdsgv02 ~]# systemctl stop postgresql-17.service
[root@lxicbpgdsgv02 ~]# ps -ef | grep postgres
root          4269      3222  0 01:14 pts/0    00:00:00 grep --color=auto
postgres
[root@lxicbpgdsgv02 ~]#
```

6.4. Clean Data, WAL, and Archive Directories

```
# Please ensure a backup is taken, if needed, before deleting the contents.
# Remove $PGDATA contents
[postgres@lxicbpgdsgv02 ~]$ rm -rf /pgdata/pgsql17/data/*
[postgres@lxicbpgdsgv02 ~]$ ls -ltr /pgdata/pgsql17/data
total 0
[postgres@lxicbpgdsgv02 ~]$

# Remove WAL log files (Redo log files)
[postgres@lxicbpgdsgv02 ~]$ rm -rf /pgwal/pgsql17/wal/*
[postgres@lxicbpgdsgv02 ~]$ ls -ltr /pgwal/pgsql17/wal
total 0
[postgres@lxicbpgdsgv02 ~]$

# Remove WAL Archive log files
[postgres@lxicbpgdsgv02 ~]$ rm -rf /pgarchive/pgsql17/archive/*
[postgres@lxicbpgdsgv02 ~]$ ls -ltr /pgarchive/pgsql17/archive
total 0
[postgres@lxicbpgdsgv02 ~]$
```

7. Restore Backup on Target

7.1. Extract Backup Files

```
[postgres@lxicbpgdsgv02 ~]$ cd /pgbackup/pgsql17/backup
[postgres@lxicbpgdsgv02 backup]$ ls -ltr
total 618052
-rw-r--r--. 1 postgres postgres 67123200 Oct 12 01:04
archive_bkp110CT2025.tar
```

```

-rw-r--r--. 1 postgres postgres 565760000 Oct 12 01:04
base_bkp11OCT2025.tar
[postgres@lxicbpgdsgv02 backup]$
[postgres@lxicbpgdsgv02 backup]$ tar -xvf base_bkp11OCT2025.tar
base_bkp11OCT2025/
base_bkp11OCT2025/16542.tar
base_bkp11OCT2025/pg_wal.tar
base_bkp11OCT2025/16545.tar
base_bkp11OCT2025/base.tar
base_bkp11OCT2025/backup_manifest
[postgres@lxicbpgdsgv02 backup]$
[postgres@lxicbpgdsgv02 backup]$ tar -xvf archive_bkp11OCT2025.tar
archive_bkp11OCT2025/
archive_bkp11OCT2025/0000000100000000000000008A
archive_bkp11OCT2025/0000000100000000000000008B
archive_bkp11OCT2025/0000000100000000000000008C
archive_bkp11OCT2025/0000000100000000000000008C.00000028.backup
archive_bkp11OCT2025/0000000100000000000000008D
[postgres@lxicbpgdsgv02 backup]$
[postgres@lxicbpgdsgv02 backup]$ ls -ltr
total 618052
drwxr-xr-x. 2 postgres postgres      97 Oct 12 00:42 base_bkp11OCT2025
drwxr-xr-x. 2 postgres postgres     182 Oct 12 00:56 archive_bkp11OCT2025
-rw-r--r--. 1 postgres postgres 67123200 Oct 12 01:04
archive_bkp11OCT2025.tar
-rw-r--r--. 1 postgres postgres 565760000 Oct 12 01:04
base_bkp11OCT2025.tar
[postgres@lxicbpgdsgv02 backup]$
[postgres@lxicbpgdsgv02 backup]$ cd base_bkp11OCT2025
[postgres@lxicbpgdsgv02 base_bkp11OCT2025]$ ls -ltr
total 552504
-rw-----. 1 postgres postgres 7940096 Oct 12 00:42 16542.tar
-rw-----. 1 postgres postgres 7905792 Oct 12 00:42 16545.tar
-rw-----. 1 postgres postgres 16778752 Oct 12 00:42 pg_wal.tar
-rw-----. 1 postgres postgres 532883456 Oct 12 00:42 base.tar
-rw-----. 1 postgres postgres 243611 Oct 12 00:42 backup_manifest
[postgres@lxicbpgdsgv02 base_bkp11OCT2025]$

```

7.2. Restore Base Data Directory

```

# Restore base backup to $PGDATA

[postgres@lxicbpgdsgv02 ~]$ ls -ltr /pgdata/pgsql17/data/
total 0

```

```

[postgres@lxicbpgdsgv02 ~]$ nohup tar -xvf
/pgbackup/pgsql17/backup/base_bkp11OCT2025/base.tar -C /pgdata/pgsql17/data
> /pgbackup/pgsql17/backup/base_restore.log 2>&1 &
[1] 4482
[postgres@lxicbpgdsgv02 ~]$

[postgres@lxicbpgdsgv02 ~]$ ls -ltr /pgdata/pgsql17/data/
total 80
-rw-----. 1 postgres postgres 227 Oct 10 18:41 backup_label.old
-rw-----. 1 postgres postgres 1169 Oct 10 18:41
postgresql.conf.bkp_10sep2025
-rw-----. 1 postgres postgres 30702 Oct 10 18:41 postgresql.conf.bkp
-rw-----. 1 postgres postgres 88 Oct 10 18:41 postgresql.auto.conf
drwx-----. 2 postgres postgres 18 Oct 10 18:41 pg_xact
-rw-----. 1 postgres postgres 3 Oct 10 18:41 PG_VERSION
drwx-----. 2 postgres postgres 6 Oct 10 18:41 pg_twophase
drwx-----. 2 postgres postgres 6 Oct 10 18:41 pg_stat_tmp
drwx-----. 2 postgres postgres 6 Oct 10 18:41 pg_snapshots
drwx-----. 2 postgres postgres 6 Oct 10 18:41 pg_serial
drwx-----. 2 postgres postgres 6 Oct 10 18:41 pg_notify
drwx-----. 4 postgres postgres 36 Oct 10 18:41 pg_multixact
-rw-----. 1 postgres postgres 2640 Oct 10 18:41 pg_ident.conf
-rw-----. 1 postgres postgres 5600 Oct 10 18:41 pg_hba.conf
drwx-----. 2 postgres postgres 6 Oct 10 18:41 pg_dynshmem
drwx-----. 2 postgres postgres 6 Oct 10 18:41 pg_commit_ts
drwx-----. 2 postgres postgres 6 Oct 10 19:45 pg_subtrans
-rw-----. 1 postgres postgres 1169 Oct 11 02:42 postgresql.conf
drwx-----. 2 postgres postgres 6 Oct 11 02:57 pg_tblspc
drwx-----. 5 postgres postgres 33 Oct 11 03:01 base
drwx-----. 2 postgres postgres 6 Oct 12 00:42 pg_replslot
drwx-----. 4 postgres postgres 68 Oct 12 00:42 pg_logical
-rw-----. 1 postgres postgres 68 Oct 12 00:42 tablespace_map
-rw-----. 1 postgres postgres 227 Oct 12 00:42 backup_label
drwx-----. 4 postgres postgres 45 Oct 12 01:29 pg_wal <-- it created as
directory.
drwx-----. 2 postgres postgres 4096 Oct 12 01:29 global
drwx-----. 2 postgres postgres 6 Oct 12 2025 pg_stat
drwx-----. 2 postgres postgres 162 Oct 12 2025 log
-rw-----. 1 postgres postgres 30 Oct 12 2025 current_logfiles
[postgres@lxicbpgdsgv02 ~]$

```

7.3. Restore WAL Directory

```
# Restore WAL Files to New WAL location

[postgres@lxicbpgdsgv02 ~]$ ls -ltr /pgwal/pgsql17/wal/
total 0
[postgres@lxicbpgdsgv02 ~]$ nohup tar -xvf
/pgbackup/pgsql17/backup/base_bkp11OCT2025/pg_wal.tar -C /pgwal/pgsql17/wal
> /pgbackup/pgsql17/backup/wal_restore.log 2>&1 &
[1] 4495
[postgres@lxicbpgdsgv02 ~]$
[postgres@lxicbpgdsgv02 ~]$ ls -ltr /pgwal/pgsql17/wal/
total 16384
-rw-----. 1 postgres postgres 16777216 Oct 12 00:42
0000000100000000000000008C
[postgres@lxicbpgdsgv02 ~]$
```

7.4. Remove and Symlink pg_wal

```
# Remove default pg_wal directory from $PGDATA
# Create symbolic link to new WAL file location
[postgres@lxicbpgdsgv02 ~]$ rm -rf /pgdata/pgsql17/data/pg_wal
[postgres@lxicbpgdsgv02 ~]$
[postgres@lxicbpgdsgv02 ~]$ ln -s /pgwal/pgsql17/wal
/pgdata/pgsql17/data/pg_wal
[postgres@lxicbpgdsgv02 ~]$
[postgres@lxicbpgdsgv02 ~]$ ls -ld /pgdata/pgsql17/data/pg_wal
lrwxrwxrwx. 1 postgres postgres 18 Oct 12 01:50 /pgdata/pgsql17/data/pg_wal
-> /pgwal/pgsql17/wal
[postgres@lxicbpgdsgv02 ~]$
```

7.5. Update \$PGDATA/tablespace_map

```
[postgres@lxicbpgdsgv02 ~]$ cat /pgdata/pgsql17/data/tablespace_map
16542 /pgTb/pgsql17/tbs/dell1tbs01
16545 /pgTb/pgsql17/tbs/orcl1tbs01
[postgres@lxicbpgdsgv02 ~]$

# Update to new location
[postgres@lxicbpgdsgv02 ~]$ sed -i 's|/pgTb/|/pgtsdata01/|g'
/pgdata/pgsql17/data/tablespace_map
[postgres@lxicbpgdsgv02 ~]$
[postgres@lxicbpgdsgv02 ~]$ cat /pgdata/pgsql17/data/tablespace_map
16542 /pgtsdata01/pgsql17/tbs/dell1tbs01
```

```
16545 /pgtsdata01/pgsql17/tbs/orcltbs01
[postgres@lxicbpgdsgv02 ~]$
[postgres@lxicbpgdsgv02 ~]$ ls -ltr /pgtsdata01/pgsql17/tbs/delltbs01
total 0
[postgres@lxicbpgdsgv02 ~]$ ls -ltr /pgtsdata01/pgsql17/tbs/orcltbs01
total 0
[postgres@lxicbpgdsgv02 ~]$
```

7.6. Restore Tablespace Data

```
# Restore Tablespace DELL_TB_DATA
[postgres@lxicbpgdsgv02 ~]$ nohup tar -xvf
/pgbackup/pgsql17/backup/base_bkp11OCT2025/16542.tar -C
/pgtsdata01/pgsql17/tbs/delltbs01 >
/pgbackup/pgsql17/backup/16542_restore.log 2>&1 &
[1] 4593
[postgres@lxicbpgdsgv02 ~]$

[postgres@lxicbpgdsgv02 ~]$ ls -ltr /pgtsdata01/pgsql17/tbs/delltbs01
total 0
drwx-----. 3 postgres postgres 19 Oct 11 02:55 PG_17_202406281 <---
[postgres@lxicbpgdsgv02 ~]$

# Restore Tablespace ORCL_TB_DATA
[postgres@lxicbpgdsgv02 ~]$ nohup tar -xvf
/pgbackup/pgsql17/backup/base_bkp11OCT2025/16545.tar -C
/pgtsdata01/pgsql17/tbs/orcltbs01 >
/pgbackup/pgsql17/backup/16545_restore.log 2>&1 &
[1] 4594
[postgres@lxicbpgdsgv02 ~]$

[postgres@lxicbpgdsgv02 ~]$ ls -ltr /pgtsdata01/pgsql17/tbs/orcltbs01
total 0
drwx-----. 3 postgres postgres 19 Oct 11 02:58 PG_17_202406281 <---
[postgres@lxicbpgdsgv02 ~]$
```

7.7. Restore WAL Archive files to backup location

```
# I have already extracted Archivelog files in the step number 7.1
# Please use this location for restore_command in postgresql.conf file

[postgres@lxicbpgdsgv02 ~]$ cd
/pgbackup/pgsql17/backup/archive_bkp11OCT2025/
```

```
[postgres@lxicbpgdsgv02 archive_bkp11OCT2025]$ ls -ltr
total 65540
-rw-----. 1 postgres postgres 16777216 Oct 12 00:53
0000000100000000000000008D
-rw-----. 1 postgres postgres      341 Oct 12 00:53
000000010000000000000008C.00000028.backup
-rw-----. 1 postgres postgres 16777216 Oct 12 00:53
0000000100000000000000008C
-rw-----. 1 postgres postgres 16777216 Oct 12 00:53
0000000100000000000000008B
-rw-----. 1 postgres postgres 16777216 Oct 12 00:53
0000000100000000000000008A
[postgres@lxicbpgdsgv02 archive_bkp11OCT2025]$
```

7.8. Create \$PGDATA/recovery.signal file

```
[postgres@lxicbpgdsgv02 ~]$ ls -ltr /pgdata/pgsql17/data/recovery.signal
ls: cannot access '/pgdata/pgsql17/data/recovery.signal': No such file or
directory
[postgres@lxicbpgdsgv02 ~]$
[postgres@lxicbpgdsgv02 ~]$ touch /pgdata/pgsql17/data/recovery.signal
[postgres@lxicbpgdsgv02 ~]$
[postgres@lxicbpgdsgv02 ~]$ ls -ltr /pgdata/pgsql17/data/recovery.signal
-rw-r--r--. 1 postgres postgres 0 Oct 12 02:06
/pgdata/pgsql17/data/recovery.signal
[postgres@lxicbpgdsgv02 ~]$
[postgres@lxicbpgdsgv02 ~]$ chmod 700 /pgdata/pgsql17/data/recovery.signal
[postgres@lxicbpgdsgv02 ~]$
```

7.9. Set Permissions

```
[postgres@lxicbpgdsgv02 ~]$ chown -R postgres:postgres /pgdata/pgsql17/data
[postgres@lxicbpgdsgv02 ~]$ chown -R postgres:postgres /pgwal/pgsql17/wal
[postgres@lxicbpgdsgv02 ~]$ chown -R postgres:postgres
/pgarchive/pgsql17/archive
[postgres@lxicbpgdsgv02 ~]$
[postgres@lxicbpgdsgv02 ~]$ chmod 700 /pgdata/pgsql17/data
[postgres@lxicbpgdsgv02 ~]$ chmod 700 /pgwal/pgsql17/wal
[postgres@lxicbpgdsgv02 ~]$ chmod 700 /pgarchive/pgsql17/archive
[postgres@lxicbpgdsgv02 ~]$
```

8. PostgreSQL Configuration on Target

8.1. Update postgresql.conf

```
# • restore_command
# • archive_mode (Optional)
# • archive_command (Optional)

[postgres@lxicbpgdsgv02 ~]$ cat /pgdata/pgsql17/data/postgresql.conf |
egrep -i 'archive_|command'
archive_mode = on
archive_command = 'cp %p /pgArch/pgsql17/arch/%f'
[postgres@lxicbpgdsgv02 ~]$

# Update archive_mode=on (Optional)
# Update archive_command='cp %p /pgarchive/pgsql17/archive/%f' (Optional)
# Update restore_command = 'cp
/pgbackup/pgsql17/backup/archive_bkp11OCT2025/%f %p' <--- Must required
for full recovery using WAL Archive log files.

[postgres@lxicbpgdsgv02 ~]$ cat /pgdata/pgsql17/data/postgresql.conf |
egrep -i 'archive_|command'
archive_mode = on
archive_command='cp %p /pgarchive/pgsql17/archive/%f'
restore_command = 'cp /pgbackup/pgsql17/backup/archive_bkp11OCT2025/%f %p'
<----- must be set correctly
[postgres@lxicbpgdsgv02 ~]$
```

8.2. Validate PGDATA in systemd and bash shell profile

```
[postgres@lxicbpgdsgv02 ~]$ cat /usr/lib/systemd/system/postgresql-
17.service | grep -i "Environment=PGDATA"
Environment=PGDATA=/pgdata/pgsql17/data
[postgres@lxicbpgdsgv02 ~]$
[postgres@lxicbpgdsgv02 ~]$ cat .bash_profile | grep -i PGDATA=
PGDATA=/pgdata/pgsql17/data
[postgres@lxicbpgdsgv02 ~]$
```

9. Start PostgreSQL Service

```
[root@lxicbpgdsgv02 ~]# systemctl start postgresql-17.service
[root@lxicbpgdsgv02 ~]#
[root@lxicbpgdsgv02 ~]# systemctl status postgresql-17.service
```

- postgresql-17.service - PostgreSQL 17 database server
 Loaded: loaded (/usr/lib/systemd/system/postgresql-17.service; enabled; preset: disabled)
 Active: active (running) since Sun 2025-10-12 02:23:36 +08; 1min 17s ago
 Docs: <https://www.postgresql.org/docs/17/static/>
 Process: 4887 ExecStartPre=/usr/pgsql-17/bin/postgresql-17-check-db-dir \${PGDATA} (code=exited, status=0/SUCCESS)
 Main PID: 4892 (postgres)
 Tasks: 8 (limit: 20496)
 Memory: 66.3M
 CPU: 200ms
 CGroup: /system.slice/postgresql-17.service
 ├─4892 /usr/pgsql-17/bin/postgres -D /pgdata/pgsql17/data
 ├─4893 "postgres: logger "
 ├─4894 "postgres: checkpointer "
 ├─4895 "postgres: background writer "
 ├─4905 "postgres: walwriter "
 ├─4906 "postgres: autovacuum launcher "
 ├─4907 "postgres: archiver last was 00000002.history"
 └─4908 "postgres: logical replication launcher "

```

Oct 12 02:23:35 lxicbpgdsgv02.rajasekhar.com systemd[1]: Starting
PostgreSQL 17 database server...
Oct 12 02:23:35 lxicbpgdsgv02.rajasekhar.com postgres[4892]: 2025-10-12
02:23:35.389 +08 [4892] LOG:  redirecting log output to logging collector
process
Oct 12 02:23:35 lxicbpgdsgv02.rajasekhar.com postgres[4892]: 2025-10-12
02:23:35.389 +08 [4892] HINT:  Future log output will appear in directory
"log".
Oct 12 02:23:36 lxicbpgdsgv02.rajasekhar.com systemd[1]: Started PostgreSQL
17 database server.
[root@lxicbpgdsgv02 ~]#

```

10. Final Validation

10.1. Check Database and Tablespace Listing

```

[postgres@lxicbpgdsgv02 ~]$ psql
psql (17.6)
Type "help" for help.

postgres=# \l+

```

List of databases

Name	Owner	Encoding	Locale Provider	Collate	Ctype	Locale	ICU Rules	Access privileges	Size	Tablespace	Description
dell	postgres	UTF8	libc	en_SG.UTF-8	en_SG.UTF-8				7755 kB	dell_tb_data	
orcl	postgres	UTF8	libc	en_SG.UTF-8	en_SG.UTF-8				7569 kB	orcl_tb_data	
postgres	postgres	UTF8	libc	en_SG.UTF-8	en_SG.UTF-8				492 MB	pg_default	default administrative connection database
template0	postgres	UTF8	libc	en_SG.UTF-8	en_SG.UTF-8				7545 kB	pg_default	unmodifiable empty database
=c/postgres											
postgres=CTc/postgres											
template1	postgres	UTF8	libc	en_SG.UTF-8	en_SG.UTF-8				7569 kB	pg_default	default template for new databases
=c/postgres											
postgres=CTc/postgres											

(5 rows)

postgres=# \db

List of tablespaces

Name	Owner	Location
dell_tb_data	postgres	/pgtsdata01/pgsql17/tbs/delltbs01
orcl_tb_data	postgres	/pgtsdata01/pgsql17/tbs/orcltbs01
pg_default	postgres	
pg_global	postgres	

(4 rows)

postgres=#

10.2. Verify Application Data

```
[postgres@lxicbpgdsgv02 ~]$ psql
psql (17.6)
Type "help" for help.
```

```
postgres=# \c dell
```

You are now connected to database "dell" as user "postgres".

```
dell=# select * from test.emp;
```

name	designation	project	company
------	-------------	---------	---------

```

-----+-----+-----+-----
Sugi | DBA | Jetstar | iGATE
Teja | DBA | RCM | iGATE
RAJ | DBA | RCM | iGATE
KK | DBA LEAD | RCM | iGATE <----
(4 rows)

```

```

dell=# exit
[postgres@lxicbpgdsgv02 ~]$

```

10.3. Validate Symbolic Links for Tablespaces

```

[postgres@lxicbpgdsgv02 ~]$ ls -ltr /pgdata/pgsql17/data/pg_tblspc/
total 0
lrwxrwxrwx. 1 postgres postgres 33 Oct 12 02:23 16545 ->
/pgtsdata01/pgsql17/tbs/orcltbs01
lrwxrwxrwx. 1 postgres postgres 33 Oct 12 02:23 16542 ->
/pgtsdata01/pgsql17/tbs/delltbs01
[postgres@lxicbpgdsgv02 ~]$

```

10.4. Confirm recovery.signal Auto-deletion

```

# File should be gone after successful recovery.

[postgres@lxicbpgdsgv02 ~]$ ls -ltr /pgdata/pgsql17/data/recovery.signal
ls: cannot access '/pgdata/pgsql17/data/recovery.signal': No such file or
directory
[postgres@lxicbpgdsgv02 ~]$

```

10.5. Review PostgreSQL Logs

```

[postgres@lxicbpgdsgv02 ~]$ cat /pgdata/pgsql17/data/log/postgresql-Sun.log
...
...
2025-10-12 02:23:35.389 +08 [4892] LOG:  starting PostgreSQL 17.6 on
x86_64-pc-linux-gnu, compiled by gcc (GCC) 11.5.0 20240719 (Red Hat 11.5.0-
5), 64-bit
2025-10-12 02:23:35.391 +08 [4892] LOG:  listening on IPv6 address ":::1",
port 5432

```

```
2025-10-12 02:23:35.392 +08 [4892] LOG:  listening on IPv4 address
"127.0.0.1", port 5432
2025-10-12 02:23:35.409 +08 [4892] LOG:  listening on Unix socket
"/run/postgresql/.s.PGSQL.5432"
2025-10-12 02:23:35.418 +08 [4892] LOG:  listening on Unix socket
"/tmp/.s.PGSQL.5432"
2025-10-12 02:23:35.424 +08 [4896] LOG:  database system was interrupted;
last known up at 2025-10-12 00:42:13 +08
2025-10-12 02:23:35.424 +08 [4896] LOG:  creating missing WAL directory
"pg_wal/archive_status"
2025-10-12 02:23:35.424 +08 [4896] LOG:  creating missing WAL directory
"pg_wal/summaries"
cp: cannot stat
'/pgbackup/pgsql17/backup/archive_bkp11OCT2025/00000002.history': No such
file or directory
2025-10-12 02:23:36.418 +08 [4896] LOG:  starting backup recovery with redo
LSN 0/8C000028, checkpoint LSN 0/8C000080, on timeline ID 1
2025-10-12 02:23:36.440 +08 [4896] LOG:  restored log file
"000000010000000000000008C" from archive
2025-10-12 02:23:36.505 +08 [4896] LOG:  starting archive recovery
2025-10-12 02:23:36.516 +08 [4896] LOG:  redo starts at 0/8C000028
2025-10-12 02:23:36.539 +08 [4896] LOG:  restored log file
"000000010000000000000008D" from archive
cp: cannot stat
'/pgbackup/pgsql17/backup/archive_bkp11OCT2025/000000010000000000000008E':
No such file or directory
2025-10-12 02:23:36.597 +08 [4896] LOG:  completed backup recovery with
redo LSN 0/8C000028 and end LSN 0/8C000120
2025-10-12 02:23:36.597 +08 [4896] LOG:  consistent recovery state reached
at 0/8C000120
2025-10-12 02:23:36.599 +08 [4892] LOG:  database system is ready to accept
read-only connections
cp: cannot stat
'/pgbackup/pgsql17/backup/archive_bkp11OCT2025/000000010000000000000008E':
No such file or directory
2025-10-12 02:23:36.612 +08 [4896] LOG:  redo done at 0/8D000310 system
usage: CPU: user: 0.00 s, system: 0.00 s, elapsed: 0.09 s
2025-10-12 02:23:36.612 +08 [4896] LOG:  last completed transaction was at
log time 2025-10-12 00:52:29.813112+08
2025-10-12 02:23:36.639 +08 [4896] LOG:  restored log file
"000000010000000000000008D" from archive
cp: cannot stat
'/pgbackup/pgsql17/backup/archive_bkp11OCT2025/00000002.history': No such
file or directory
2025-10-12 02:23:36.700 +08 [4896] LOG:  selected new timeline ID: 2
cp: cannot stat
```

```
'/pgbackup/pgsql17/backup/archive_bkp11OCT2025/00000001.history': No such
file or directory
2025-10-12 02:23:36.779 +08 [4896] LOG:  archive recovery complete
2025-10-12 02:23:36.782 +08 [4894] LOG:  checkpoint starting: end-of-
recovery immediate wait
2025-10-12 02:23:36.810 +08 [4894] LOG:  checkpoint complete: wrote 4
buffers (0.0%); 0 WAL file(s) added, 0 removed, 2 recycled; write=0.002 s,
sync=0.008 s, total=0.031 s; sync files=3, longest=0.003 s, average=0.003
s; distance=32768 kB, estimate=32768 kB; lsn=0/8E000028, redo
lsn=0/8E000028
2025-10-12 02:23:36.823 +08 [4892] LOG:  database system is ready to accept
connections
[postgres@lxicbpgdsgv02 ~]$
```

10.6. Check DB recovery status

```
[postgres@lxicbpgdsgv02 ~]$ psql -c "SELECT pg_is_in_recovery();"
pg_is_in_recovery
-----
f <--- means recovery is completed.
(1 row)

[postgres@lxicbpgdsgv02 ~]$
```

Caution: *Your use of any information or materials on this website is entirely at your own risk. It is provided for educational purposes only. It has been tested internally, however, we do not guarantee that it will work for you. Ensure that you run it in your test environment before using.*

Thank you,

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